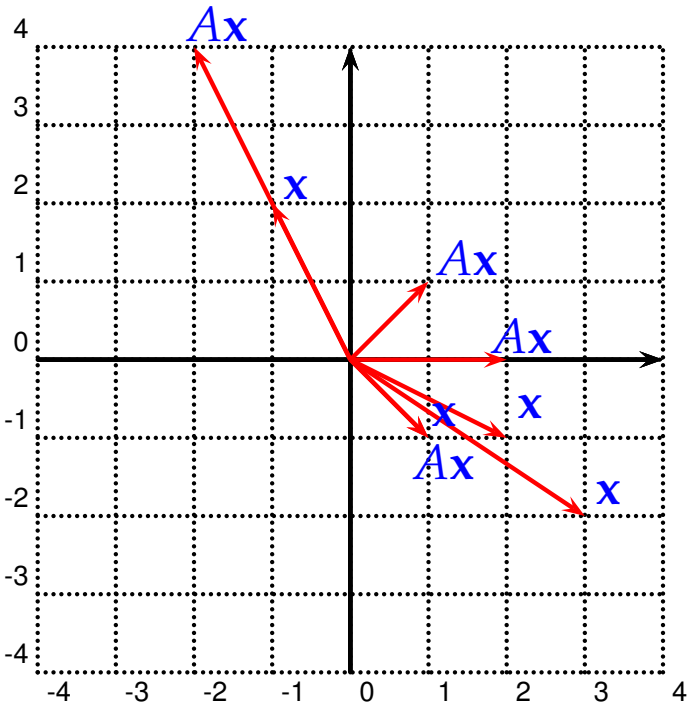
Recall : Eigenvalues and Eigenvectors

Let us start with a simple example.

Problem. Consider the matrix $A = \begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}$. What does multiplication by A do to vectors in $\mathbb{R}^2$?

$$\mathbf{x} \mapsto A\mathbf{x}$$



$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Recall : Eigenvalues and Eigenvectors cont.)

Let A be an $n \times n$ matrix. If a nonzero vector $\mathbf{x}$ in $\mathbb{R}^n$ satisfies $A\mathbf{x} = \lambda\mathbf{x}$ for a scalar λ, then :

- The scalar λ is called an eigenvalue of A.
- The vector $\mathbf{x}$ is called an eigenvector of A corresponding to λ.

Example. $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}$ corresponding to $\lambda = 2$ because

$$\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

The eigenvector must be nonzero, but the eigenvalue can be zero or nonzero.

Examples

Example. The vectors $\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}-2\\-2\end{bmatrix}, \begin{bmatrix}\sqrt{3}\\\sqrt{3}\end{bmatrix}$ are eigenvectors of $\begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}$ corresponding to $\lambda = 3$, because :

$$\begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}3\\3\end{bmatrix} = 3\begin{bmatrix}1\\1\end{bmatrix}$$

$$\begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}\begin{bmatrix}-2\\-2\end{bmatrix} = \begin{bmatrix}-6\\-6\end{bmatrix} = 3\begin{bmatrix}-2\\-2\end{bmatrix}$$

$$\begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}\begin{bmatrix}\sqrt{3}\\\sqrt{3}\end{bmatrix} = \begin{bmatrix}3\sqrt{3}\\3\sqrt{3}\end{bmatrix} = 3\begin{bmatrix}\sqrt{3}\\\sqrt{3}\end{bmatrix}$$

If $\mathbf{x}$ is an eigenvector of A corresponding to λ, then for every scalar $c \neq 0$, the vector $c\mathbf{x}$ is also an eigenvector of A with eigenvalue λ.

Reason : $A(c\mathbf{x}) = c(A\mathbf{x}) = c\lambda\mathbf{x} = \lambda(c\mathbf{x})$.

How can we check if a given λ is an eigenvalue of A?

Let A be an $n \times n$ matrix.

- A scalar λ is an eigenvalue of A if and only if the homogeneous linear system

$$(A - \lambda \mathrm{I})\mathbf{x} = \mathbf{0}$$

has nontrivial solutions.

- Here "I" denotes the $n \times n$ identity matrix.

Eigenspaces

Let A be an $n \times n$ matrix.

If λ is an eigenvalue of A, the set of vectors $\mathbf{x}$ in $\mathbb{R}^n$ satisfying

$$A\mathbf{x} = \lambda\mathbf{x}$$

forms a subspace of $\mathbb{R}^n$. It is called the eigenspace of A corresponding to λ.

- Indeed the eigenspace corresponding to λ is identical to $\mathrm{Nul}(A - \lambda \mathrm{I})$

$$A\mathbf{x} = \lambda\mathbf{x} \Leftrightarrow (A - \lambda \mathrm{I})\mathbf{x} = \mathbf{0}$$

Example

Is -5 an eigenvalue of $A = \begin{bmatrix} -4 & -1 & 2 \\ -1 & -4 & -2 \\ 2 & -2 & -1 \end{bmatrix}$? If it is, then find a basis for the eigenspace corresponding to -5.

Solution.

$A\mathbf{x} = -5\mathbf{x} \Rightarrow A\mathbf{x} + 5\mathbf{x} = \mathbf{0} \Rightarrow (A + 5\mathrm{I})\mathbf{x} = 0$

$$A + 5\mathrm{I} = \begin{bmatrix} -4 & -1 & 2 \\ -1 & -4 & -2 \\ 2 & -2 & -1 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix}$$

We want to solve $(A + 5\mathrm{I})\mathbf{x} = \mathbf{0}$, i.e., $\begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ -1 & 1 & -2 & 0 \\ 2 & -2 & 4 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \{\, x_1 - x_2 + 2x_3 = 0$$

x_1 : basic, x_2, x_3 : free.

The homogeneous system has nontrivial solutions $\Rightarrow -5$ is an eigenvalue of A.

$$x_1 = x_2 - 2x_3 \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 - 2x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

basis for eigenspace: $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}$

Example

Is -5 an eigenvalue of $A = \begin{bmatrix} -4 & -1 & 2 \\ -1 & -4 & -2 \\ 2 & -2 & -1 \end{bmatrix}$? If it is, then find a basis for the eigenspace corresponding to -5.

Solution.

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\} \text{ is a basis for the eigenspace corresponding to } -5.$$

Note that you can also check your answer :

$$\begin{bmatrix} -4 & -1 & 2 \\ -1 & -4 & -2 \\ 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -5 \\ -5 \\ 0 \end{bmatrix} = (-5) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -4 & -1 & 2 \\ -1 & -4 & -2 \\ 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ -5 \end{bmatrix} = (-5) \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

Eigenvalues of a triangular matrix

Theorem

The eigenvalues of a triangular matrix are the diagonal entries of the matrix.

Example. The eigenvalues of $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -2 & 2 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 \\ 4 & 1 & 5 & 2 & 0 \\ 0 & -3 & 4 & 5 & -1 \end{bmatrix}$ are -1, 0, 1, 2.

Theorem

Let A be a square matrix whose eigenvalues are $\lambda_1, ..., \lambda_k$. Then the eigenvalues of A^r are $\lambda_1^r, ..., \lambda_k^r$.

Theorem

Let A be a square matrix whose eigenvalues are $\lambda_1, ..., \lambda_k$. Then the eigenvalues of A^r are $\lambda_1^r, ..., \lambda_k^r$.

Example

If $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 & 0 \\ 2 & -1 & 0 & 0 & 0 \\ 4 & 1 & 5 & 2 & 0 \\ 0 & 5 & 4 & 5 & -1 \end{bmatrix}$, what are the eigenvalues of A^3 and A^4?

Solution.

- Since A is triangular, the eigenvalues of A are its diagonal entries, i.e., -1, 0, 1, 2.
- The eigenvalues of A^3 are $(-1)^3, 0^3, 1^3, 2^3$; i.e., -1, 0, 1, 8.
- The eigenvalues of A^4 are $(-1)^4, 0^4, 1^4, 2^4$; i.e., 0, 1, 16.

Reminder

Suppose a, b, c are given numbers.
How do we solve the equation $ax^2 + bx + c = 0$ for x?

Solution.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Finding Eigenvalues

Example

Find eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$.

Suppose $\mathbf{x}$ is a nonzero vector in $\mathbb{R}^2$ such that $A\mathbf{x} = \lambda\mathbf{x}$. Then :

$$A\mathbf{x} = \lambda\mathbf{x} \quad \Longrightarrow \quad A\mathbf{x} - \lambda\mathbf{x} = \mathbf{0} \quad \Longrightarrow \quad (A - \lambda \mathrm{I})\mathbf{x} = \mathbf{0} \quad \text{where } \mathrm{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$\mathrm{Nul}(A - \lambda \mathrm{I}) \neq \{\, \mathbf{0} \,\} \quad \Longrightarrow \quad A - \lambda \mathrm{I} \text{ is not invertible} \quad \Longrightarrow \quad \det(A - \lambda \mathrm{I}) = 0.$$

$$A - \lambda \mathrm{I} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 1 - \lambda & 3 \\ 3 & 1 - \lambda \end{bmatrix}$$

$$\det(A - \lambda \mathrm{I}) = (1 - \lambda)^2 - 9 = \lambda^2 - 2\lambda - 8$$

$$\det(A - \lambda \mathrm{I}) = 0 \quad \Longrightarrow \quad \lambda^2 - 2\lambda - 8 = 0$$

$$\Longrightarrow \quad \lambda = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(-8)}}{2} = \begin{cases} -2 \\ 4 \end{cases}$$

The eigenvalues of A are $\lambda = -2$ and $\lambda = 4$.

Finding Eigenvalues and Eigenvectors (Cont.)

Example

Find eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$.

$\boxed{\lambda = -2:}$ We need to find $\text{Nul}(A - (-2)\text{I})$.

$$A - (-2)\text{I} = A + 2\text{I} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$$

$\text{Nul}(A + 2\text{I})$ is the solution set of the homogeneous system $\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\left[\begin{array}{cc|c} 3 & 3 & 0 \\ 3 & 3 & 0 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{3}R_1} \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 3 & 3 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$\implies \quad \{x_1 + x_2 = 0 \ \Rightarrow \ x_1 + x_2 = 0 \quad \implies \quad x_1 = -x_2 \qquad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Any vector of the form $c \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, with $c \neq 0$, is an eigenvector of $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$ with $\lambda = -2$.

One can check that $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix} = (-2) \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Finding Eigenvalues and Eigenvectors (Cont.)

Example

Find eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$.

$\boxed{\lambda = 4:}$ We need to find a basis for $\text{Nul}(A - 4\text{I})$.

$$A - 4\text{I} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix}$$

$\text{Nul}(A - 4\text{I})$ is the solution set of the homogeneous system $\begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\left[\begin{array}{cc|c} -3 & 3 & 0 \\ 3 & -3 & 0 \end{array}\right] \xrightarrow{R_1 \to -\frac{1}{3}R_1} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 3 & -3 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$\implies \quad \{x_1 - x_2 = 0 \ \Rightarrow \ x_1 - x_2 = 0 \quad \implies \quad x_1 = x_2 \qquad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Any vector of the form $c \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, with $c \neq 0$, is an eigenvector of $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$ with $\lambda = 4$.

One can check that $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Finding Eigenvalues and Eigenvectors

Let A be an $n \times n$ matrix.

- To find the eigenvalues of A, we solve the equation $\det(A - \lambda \mathrm{I}) = 0$ for λ.
- For each obtained eigenvalue λ, the eigenspace corresponding to λ is identical to $\mathrm{Nul}(A - \lambda \mathrm{I})$.

The characteristic equation

Let A be an $n \times n$ matrix.

The equation $\det(A - \lambda \mathrm{I}) = 0$ is called the characteristic equation of A.

- The characteristic equation has the following form :

$$\pm\lambda^n + c_{n-1}\lambda^{n-1} + c_{n-2}\lambda^{n-2} + \cdots + c_1\lambda + c_0 = 0$$

Example

Find the eigenvalues of $A = \begin{bmatrix} -4 & 0 & 1 \\ 3 & 2 & -5 \\ -6 & 0 & 1 \end{bmatrix}$. Then determine bases for the associated eigenspaces.

Solution. First we find the characteristic equation :

$$\begin{aligned}
\det(A - \lambda \mathrm{I}) &= \begin{vmatrix} -4-\lambda & 0 & 1 \\ 3 & 2-\lambda & -5 \\ -6 & 0 & 1-\lambda \end{vmatrix} \\
&= -0\begin{vmatrix} 3 & -5 \\ -6 & 1-\lambda \end{vmatrix} + (2-\lambda)\begin{vmatrix} -4-\lambda & 1 \\ -6 & 1-\lambda \end{vmatrix} - 0\begin{vmatrix} -4-\lambda & 1 \\ 3 & -5 \end{vmatrix} \\
&= (2-\lambda)\Big((-4-\lambda)(1-\lambda) - (1)(-6)\Big) \\
&= (2-\lambda)(\lambda^2 + 3\lambda + 2) = (2-\lambda)(\lambda+1)(\lambda+2)
\end{aligned}$$

Because

$$\lambda^2 + 3\lambda + 2 = 0 \;\Rightarrow\; \lambda = \frac{-3 \pm \sqrt{9-8}}{2} = \begin{cases} -1 \\ -2 \end{cases}$$

- Therefore the eigenvalues of A are $2, -2, -1$.

Example

Find the eigenvalues of $A = \begin{bmatrix} -4 & 0 & 1 \\ 3 & 2 & -5 \\ -6 & 0 & 1 \end{bmatrix}$. Then determine bases for the associated eigenspaces.

$$\lambda = 2 \;\Rightarrow\; A - 2\mathrm{I} = \begin{bmatrix} -6 & 0 & 1 \\ 3 & 0 & -5 \\ -6 & 0 & -1 \end{bmatrix} \qquad \begin{bmatrix} -6 & 0 & 1 \\ 3 & 0 & -5 \\ -6 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -6 & 0 & 1 & 0 \\ 3 & 0 & -5 & 0 \\ -6 & 0 & -1 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \qquad \Rightarrow \qquad \text{basis for eigenspace of } \lambda = 2\text{: } \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Example

Find the eigenvalues of $A = \begin{bmatrix} -4 & 0 & 1 \\ 3 & 2 & -5 \\ -6 & 0 & 1 \end{bmatrix}$. Then determine bases for the associated eigenspaces.

$$\lambda = -2 \;\Rightarrow\; A - (-2)\mathrm{I} = \begin{bmatrix} -2 & 0 & 1 \\ 3 & 4 & -5 \\ -6 & 0 & 3 \end{bmatrix} \qquad \begin{bmatrix} -2 & 0 & 1 \\ 3 & 4 & -5 \\ -6 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -2 & 0 & 1 & 0 \\ 3 & 4 & -5 & 0 \\ -6 & 0 & 3 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{7}{8} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{1}{2} \\ \frac{7}{8} \\ 1 \end{bmatrix} \qquad \Rightarrow \qquad \text{basis for eigenspace of } \lambda = -2\text{: } \left\{ \begin{bmatrix} \frac{1}{2} \\ \frac{7}{8} \\ 1 \end{bmatrix} \right\}$$

Example

Find the eigenvalues of $A = \begin{bmatrix} -4 & 0 & 1 \\ 3 & 2 & -5 \\ -6 & 0 & 1 \end{bmatrix}$. Then determine bases for the associated eigenspaces.

$$\lambda = -1 \;\Rightarrow\; A - (-1)\mathrm{I} = \begin{bmatrix} -3 & 0 & 1 \\ 3 & 3 & -5 \\ -6 & 0 & 2 \end{bmatrix} \qquad \begin{bmatrix} -3 & 0 & 1 \\ 3 & 3 & -5 \\ -6 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -3 & 0 & 1 & 0 \\ 3 & 3 & -5 & 0 \\ -6 & 0 & 2 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{1}{3} & 0 \\ 0 & 1 & -\frac{4}{3} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{1}{3} \\ \frac{4}{3} \\ 1 \end{bmatrix} \qquad \Rightarrow \qquad \text{basis for eigenspace of } \lambda = -1\text{: } \left\{ \begin{bmatrix} \frac{1}{3} \\ \frac{4}{3} \\ 1 \end{bmatrix} \right\}$$

Example

Determine the characteristic polynomial of $A = \begin{bmatrix} 1 & -3 & 0 & 5 & 1 \\ 0 & 2 & 1 & -\frac{1}{2} & 7 \\ 0 & 0 & -3 & 4 & 3 \\ 0 & 0 & 0 & 3 & -5 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$

Solution.

$$A - \lambda \mathrm{I} = \begin{bmatrix} 1-\lambda & -3 & 0 & 5 & 1 \\ 0 & 2-\lambda & 1 & -\frac{1}{2} & 7 \\ 0 & 0 & -3-\lambda & 4 & 3 \\ 0 & 0 & 0 & 3-\lambda & -5 \\ 0 & 0 & 0 & 0 & 2-\lambda \end{bmatrix}$$

$$\Rightarrow \det(A-\lambda \mathrm{I}) = (1-\lambda)(2-\lambda)(-3-\lambda)(3-\lambda)(2-\lambda) = -(\lambda-1)(\lambda-2)^2(\lambda+3)(\lambda-3).$$

Theorem

Suppose A is an $n \times n$ matrix. Then the eigenvalues of A and A^T are the same.

Reason : Indeed the characteristic equations of A and A^T are the same :

$$\det(A^T - \lambda \mathrm{I}) = \det(A^T - \lambda \mathrm{I}^T) = \det\left((A - \lambda \mathrm{I})^T\right) = \det(A - \lambda \mathrm{I})$$

Example

- Eigenvalues of $A = \begin{bmatrix} 2 & -1 \\ 5 & -4 \end{bmatrix}$ are obtained as follows :

$$\det(A - \lambda \mathrm{I}) = \begin{vmatrix} 2-\lambda & -1 \\ 5 & -4-\lambda \end{vmatrix} = (2-\lambda)(-4-\lambda) - (-1)(5) = \lambda^2 + 2\lambda - 3$$

Characteristic equation of A : $\lambda^2 + 2\lambda - 3 = 0 \Rightarrow \lambda = \frac{-2 \pm \sqrt{4-(4)(-3)}}{2} = \begin{cases} -3 \\ 1 \end{cases}$

Therefore the eigenvalues of A are -3 and 1.

- Eigenvalues of A^T are obtained as follows :

$$\det(A^T - \lambda \mathrm{I}) = \begin{vmatrix} 2-\lambda & 5 \\ -1 & -4-\lambda \end{vmatrix} = (2-\lambda)(-4-\lambda) - (-1)(5) = \lambda^2 + 2\lambda - 3$$

Characteristic equation of A^T is $\lambda^2 + 2\lambda - 3 = 0$ which is the same as the characteristic equation of A.

Therefore the eigenvalues of A^T are -3 and 1 as well.

Eigenvalues of invertible matrices

Theorem

An $n \times n$ matrix A is invertible if and only if 0 is not an eigenvalue of A.

Reason : 0 is an eigenvalue of $A \Leftrightarrow \text{Nul}\,(A - 0\text{I}) \neq \{\mathbf{0}\} \Leftrightarrow A$ is not invertible.

Example

Let A be a 5×5 matrix. You don't know what A is but you know that its characteristic equation is $-\lambda^5 + 3\lambda^3 + 4\lambda^2 - \lambda = 0$. Is A invertible?

Solution. No! Because $\lambda = 0$ is a solution of the characteristic equation, so it is an eigenvalue of A. By the above theorem, A is not invertible.

Example

Let A be a 4×4 matrix. You don't know what A is but you know that its characteristic equation is $\lambda^4 - \lambda^3 + 2\lambda^2 - \lambda + 5 = 0$. Is A invertible?

Solution. Yes! Because $\lambda = 0$ is not a solution of the characteristic equation, so it is not an eigenvalue of A. By the above theorem, A is invertible.

Complex Eigenvalues

Example

Suppose $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ Find the eigenvalues and eigenvectors of A.

$$\det(A - \lambda \mathrm{I}) = \begin{vmatrix} -\lambda & 1 \\ -1 & -\lambda \end{vmatrix} = \lambda^2 + 1$$

$\lambda^2 + 1 = 0 \Rightarrow \lambda = \pm i \Rightarrow$ the eigenvalues are *not* real numbers. But that's okay!

$\boxed{\lambda = i}$

$$A - i\mathrm{I} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} - \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix} = \begin{bmatrix} -i & 1 \\ -1 & -i \end{bmatrix} \qquad \begin{bmatrix} -i & 1 \\ -1 & -i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -i & 1 & 0 \\ -1 & -i & 0 \end{array}\right] \xrightarrow{R_1 \to iR_1} \left[\begin{array}{cc|c} 1 & i & 0 \\ -1 & -i & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + R_1} \left[\begin{array}{cc|c} 1 & i & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$x_1 + ix_2 = 0 \Rightarrow x_1 = -ix_2 \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -ix_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

- Therefore the eigenspace corresponding to $\lambda = i$ is $\mathrm{Span}\left\{ \begin{bmatrix} -i \\ 1 \end{bmatrix} \right\}$.

Complex Eigenvalues (Cont.)

Example

Suppose $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ Find the eigenvalues and eigenvectors of A.

$\boxed{\lambda = -i}$

$$A - (-i)\mathrm{I} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} - \begin{bmatrix} -i & 0 \\ 0 & -i \end{bmatrix} = \begin{bmatrix} i & 1 \\ -1 & i \end{bmatrix} \qquad \begin{bmatrix} i & 1 \\ -1 & i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} i & 1 & 0 \\ -1 & i & 0 \end{array}\right] \xrightarrow{R_1 \to -iR_1} \left[\begin{array}{cc|c} 1 & -i & 0 \\ -1 & i & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + R_1} \left[\begin{array}{cc|c} 1 & -i & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$x_1 - ix_2 = 0 \Rightarrow x_1 = ix_2 \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} ix_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} i \\ 1 \end{bmatrix}$$

- Therefore the eigenspace corresponding to $\lambda = -i$ is Span $\left\{ \begin{bmatrix} i \\ 1 \end{bmatrix} \right\}$.

We can also check the answer :

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} -i \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ i \end{bmatrix} = i \begin{bmatrix} -i \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} i \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -i \end{bmatrix} = (-i) \begin{bmatrix} i \\ 1 \end{bmatrix}$$

Multiplicity of an eigenvalue

Sometimes the characteristic equation may have repeated roots.

- The number of times an eigenvalue λ appears as a root of the characteristic equation is called its multiplicity.
- If we take multiplicities into account, the characteristic equation of an $n \times n$ matrix has n roots.

Example. Suppose A is a 5×5 matrix whose characteristic equation is $-\lambda^5 - 2\lambda^4 - \lambda^3 = 0$. Find the eigenvalues of A and their multiplicities.

$$-\lambda^5 - 2\lambda^4 - \lambda^3 = -\lambda^3(\lambda^2 + 2\lambda + 1) = -\lambda^3(\lambda + 1)^2$$

Therefore the eigenvalues are :

- 0 with multiplicity 3
- -1 with mutiplicity 2

(You can see that the total sum of multiplicities is 3+2=5.)

Multiplicity and dimension of eigenspace

Theorem

Let A be a square matrix an λ an eigenvalue of A.

- In general we have :

$$\begin{array}{c}\text{Dimension of the eigenspace} \\ \text{corresponding to } \lambda\end{array} \leqslant \text{Multiplicity of } \lambda$$

- The inequality can be strict.

Example. Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. Then :

$$\det(A - \lambda \mathrm{I}) = \begin{vmatrix} 1-\lambda & 0 \\ 1 & 1-\lambda \end{vmatrix} = (1-\lambda)^2$$

Therefore $\lambda = 1$ is an eigenvalue with multiplicity 2.

However, $A\mathbf{x} = \mathbf{x} \Rightarrow (A - \mathrm{I})\mathbf{x} = 0 \Rightarrow \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$x_1 = 0 \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Which means that the eigenspace corresponding to $\lambda = 1$ is one-dimensional.